$S(t)$: stock price at time t

$\frac{dS}{S} = \mu\, dt + \sigma\, dW(t)$. $W(t)$ is a Brownian motion $[W_t - W_0 \sim N(0,t)]$

derivative $V = V(S,t)$ [with $S(t)$ as underlying asset]

$\Rightarrow$ create a risk free portfolio $\pi = \underline{V - h\cdot S}$ option stock sell h stock to hedge the stock

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1. Risk Neutral Definition

$\Delta\pi = \Delta V - h\cdot \Delta S = r\cdot(V - h\cdot S)\cdot dt$ [Return = Risk Free Return]

risk free rate; portfolio initial value; invested time

2. Ito's Lemma

$\Delta V = V(S+\Delta S, t+\Delta t) - V(S,t)$

① Taylor Expansion

$V(S+\Delta S, t+\Delta t) = V(S,t) + \left(\frac{\partial V}{\partial S}\cdot\Delta S + \frac{\partial V}{\partial t}\Delta t\right) + \frac{1}{2}\left(\frac{\partial^2 V}{\partial S^2}\cdot\Delta S^2 + \frac{\partial^2 V}{\partial t^2}\cdot\underbrace{\Delta t^2}_{\to 0} + 2\frac{\partial V}{\partial t}\frac{\partial V}{\partial x}\cdot\underbrace{\Delta S\,\Delta t}_{\to 0}\right) + \cdots$

$\Delta V = \frac{\partial V}{\partial S}\cdot\Delta S + \frac{\partial V}{\partial t}\cdot\Delta t + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\cdot\Delta S^2$

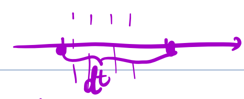
$= \frac{\partial V}{\partial S}\cdot S\cdot[\mu dt + \sigma dW(t)] + \frac{\partial V}{\partial t}\cdot\Delta t + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\cdot[S^2\cdot(\mu dt + \sigma dW(t))^2]$

$= \frac{\partial V}{\partial S}\cdot S\cdot[\mu dt + \sigma dW(t)] + \frac{\partial V}{\partial t}\cdot\Delta t + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\cdot S^2\cdot[\mu^2\underbrace{dt^2}_{\to 0} + \sigma^2\underbrace{dW^2(t)}_{dt} + 2\mu\cdot\sigma\underbrace{dt\cdot dW}_{\to 0}]$

$W_{t+s} - W_t \sim N(0,s)$

$X_i = B(t_{i+1}) - B(t_i)$

$\sum_{i=1}^{n}[B(t_{i+1}) - B(t_i)]^2$



$= \sum_{i=1}^{n} X_i^2$ $\quad X_i \sim N(0, \frac{t}{n})$

$= n\cdot\left[\frac{1}{n}\sum_{i=1}^{n} X_i^2\right]$

$= n\cdot\left[\frac{1}{n}\sum_{i=1}^{n} Y_i\right]$ $\quad$ Set $Y_i = X_i^2$

$E[Y_i] = E[X_i^2] = Var[X_i] + E^2[X_i] = \frac{dt}{n} + 0 = \frac{dt}{n}$

$= n\cdot\left(\frac{t}{n}\right) = t$

law of large number

$= \frac{\partial V}{\partial S}\cdot S[\mu dt + \sigma dW(t)]$

$+ \frac{\partial V}{\partial t}\cdot\Delta t$

$+ \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\cdot S^2\cdot\sigma^2 dt$

$= dt\left[\frac{\partial V}{\partial S}\cdot S\cdot\mu + \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\cdot S^2\cdot\sigma^2\right]$

$+ \frac{\partial V}{\partial S}\cdot S\cdot\sigma\cdot dW(t)$

$\Rightarrow \Delta V - h\cdot\Delta S = \left[\frac{\partial V}{\partial S}\cdot S\cdot\mu + \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}S^2\sigma^2\right]dt + \frac{\partial V}{\partial S}\cdot S\cdot\sigma\cdot dW(t) - h\cdot\Delta S$

$= \left[\frac{\partial V}{\partial S}\cdot S\cdot\mu + \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}S^2\sigma^2\right]dt + \frac{\partial V}{\partial S}\cdot S\cdot\sigma\cdot dW(t)$

$- h\cdot S\cdot[\mu dt + \sigma dW(t)]$

$= \left[\frac{\partial V}{\partial S}\cdot S\cdot\mu - h\cdot S\cdot\mu + \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}S^2\sigma^2\right]dt + \left(\frac{\partial V}{\partial S}\cdot S\cdot\sigma - hS\sigma\right)dW(t)$

$$= \left[\frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2} S^2 \sigma^2\right] \cancel{dt} = r \cdot \underbrace{\left[V - \frac{\partial V}{\partial S} \cdot S\right] \cancel{dt}}_{\text{risk free return}}$$

$\because$ Risk Free

$\therefore$ the return will not have the random term

$$\Rightarrow \frac{\partial V}{\partial S} \cdot s \cdot a - h s \cdot a = 0$$

$$h = \frac{\partial V}{\partial S}$$ ← Risk Neutral

$$\Rightarrow r \cdot \left[V - \frac{\partial V}{\partial S}\right] = \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2} \sigma^2 S^2.$$

$$\Rightarrow \boxed{\Rightarrow \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\sigma^2 S^2 + r \cdot \frac{\partial V}{\partial S} \cdot S - rV = 0}$$ ← Black Schole Merton Differential Equation

risk free rate

need to use $r - q$, 因为 initially short stock, 所以 did not own dividend.

dividend

$$\left[\frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2} S^2 \sigma^2\right] = \left[rV - r\frac{\partial V}{\partial S} \cdot S\right]$$

$$\Downarrow$$

$$\frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2} S^2 \sigma^2 = rV - (r-d) \cdot S \cdot \frac{\partial V}{\partial S} = 0$$

$$S = S_0 e^x \longleftarrow x = \ln\left(\frac{S}{S_0}\right) = \ln(S) - \ln(S_0)$$

log return from $S_0 \to S$

optim.

① $$\frac{\partial V}{\partial S} = \frac{\partial V}{\partial x} \cdot \frac{\partial x}{\partial S} = \frac{\partial V}{\partial x} \cdot \frac{\partial}{\partial S}\left[\ln(S) - \ln(S_0)\right] = \frac{\partial V}{\partial x} \cdot \frac{1}{S}$$

$$\Rightarrow \frac{\partial V}{\partial S} = \frac{1}{S} \cdot \frac{\partial V}{\partial x} \Rightarrow \frac{\partial V}{\partial x} = S \cdot \frac{\partial V}{\partial S}$$

$$S = S_0 e^x \longrightarrow \frac{1}{S} = \frac{1}{S_0} e^{-x}$$

$$\Rightarrow \frac{\partial}{\partial S} = \frac{1}{S} \cdot \frac{\partial}{\partial x}$$

$$\frac{\partial}{\partial x}\left(\frac{1}{S}\right) = \frac{\partial}{\partial x}\left[\frac{1}{S_0} e^{-x}\right] = -\frac{1}{S_0} e^{-x} = -\frac{1}{S}$$

② $$\frac{\partial^2 V}{\partial S^2} = \frac{\partial}{\partial S}\left[\frac{\partial V}{\partial S}\right] = \frac{\partial}{\partial S}\left[\frac{1}{S} \cdot \frac{\partial V}{\partial x}\right] = \frac{1}{S} \cdot \frac{\partial}{\partial x}\left[\frac{1}{S} \cdot \frac{\partial V}{\partial x}\right] = \frac{1}{S}\left[\frac{\partial}{\partial x}\left(\frac{1}{S}\right) \cdot \frac{\partial V}{\partial x} + \frac{1}{S}\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right)\right]$$

$$= \frac{1}{S}\left[\left(-\frac{1}{S}\right) \cdot \frac{\partial V}{\partial x} + \frac{1}{S}\frac{\partial V^2}{\partial x^2}\right]$$

$$= -\frac{1}{S^2}\frac{\partial V}{\partial x} + \frac{1}{S^2}\frac{\partial V}{\partial x^2}$$

$$S^2 \cdot \frac{\partial^2 V}{\partial S^2} = S^2\left[-\frac{1}{S^2}\frac{\partial V}{\partial x} + \frac{1}{S^2} \cdot \frac{\partial V}{\partial x^2}\right] = \frac{\partial V}{\partial x^2} - \frac{\partial V}{\partial x}$$

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2\left[\frac{\partial^2 V}{\partial x^2} - \frac{\partial V}{\partial x}\right] + (r-d) \cdot \frac{\partial V}{\partial x} - rV = 0$$

$$\frac{\partial V}{\partial t} + \frac{\partial^2 V}{\partial x^2} \cdot \frac{1}{2}\sigma^2 + \frac{\partial V}{\partial x}\left[r - d - \frac{1}{2}\sigma^2\right] - rV = 0$$

In here use two point forward difference [In BS, 2-point in time and 3 point inplace, and Backwardation]

↑ industry standard

First Derivative

Forward Difference Formula

$$f'(x) = \frac{f(x+h) - f(x)}{h}$$

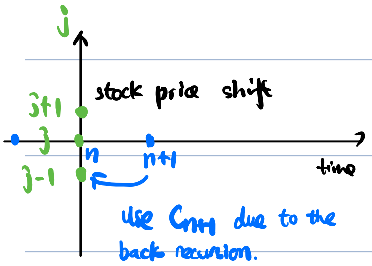
Backward Difference Formula

$$f'(x) = \frac{f(x) - f(x-h)}{h}$$

Centered Difference Formula

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

Second Derivative: $f''(x_i) = \dfrac{f(x_i) - 2f(x_{i+1}) + f(x_{i+2})}{h^2}$



$$\frac{\partial C(S,t)}{\partial t} = \frac{C^j_{n+1} - C^j_n}{\Delta t} \leftarrow \text{not symmetric difference}$$

$$\left.\begin{aligned} \frac{\partial C(S,t)}{\partial X} &= \frac{C^{j+1}_{n+1} - C^{j-1}_{n+1}}{2\Delta X} \\ \frac{\partial^2 C(S,t)}{\partial x^2} &= \frac{C^{j+1}_{n+1} + C^{j-1}_{n+1} - 2C^j_{n+1}}{(\Delta x)^2} \end{aligned}\right\} \text{symmetric difference.}$$

$$\frac{\partial V}{\partial t} + \frac{\partial^2 V}{\partial x^2}\cdot\frac{1}{2}\sigma^2 + \frac{\partial V}{\partial x}\left[r - d - \frac{1}{2}\sigma^2\right] - rV = 0$$

$$\frac{V^j_{n+1} - V^j_n}{\Delta t} + \frac{1}{2}\sigma^2\left[\frac{V^{j+1}_{n+1} + V^{j-1}_{n+1} - 2V^j_{n+1}}{(\Delta x)^2}\right] + \frac{V^{j+1}_{n+1} - V^{j-1}_{n+1}}{2\Delta x}\left[r - d - \frac{1}{2}\sigma^2\right] - rV = 0$$

$$\underbrace{V^{j+1}_{n+1}\left[\frac{1}{2(\Delta x)^2}\sigma^2 + \frac{(r-d-\frac{1}{2}\sigma^2)}{2\Delta x}\right] + V^j_{n+1}\left[\frac{1}{\Delta t} - \frac{2\sigma^2}{\Delta x^2}\right] + V^{j-1}_{n+1}\left[\frac{1}{2}\sigma^2\cdot\frac{1}{\Delta x^2} - \frac{1}{2\Delta x}(r-d-\frac{1}{2}\sigma^2)\right]}_{①} - \underbrace{\frac{V^j_n}{\Delta t} - rV}_{②} = 0$$

$\times \Delta t$

$$\left[\frac{V^j_n}{\Delta t} - rV\right]\cdot\Delta t = V^j_n - rV^j_n\cdot\Delta t = (1 - r\Delta t)V^j_n$$

$$V^{j+1}_{n+1}\underbrace{\left[\frac{1}{2}\frac{\Delta t\sigma^2}{\Delta x^2} + \frac{(r-d-\frac{1}{2}\sigma^2)}{2\Delta x}\right]}_{P_u} + V^j_{n+1}\underbrace{\left[1 - \frac{2\Delta t\sigma^2}{2\Delta x^2}\right]}_{P_m} + V^{j-1}_{n+1}\underbrace{\left[\frac{\Delta t}{2}\sigma^2\cdot\frac{1}{\Delta x^2} - \frac{\Delta t}{2\Delta x}(r-d-\frac{1}{2}\sigma^2)\right]}_{P_d}$$

$$(1 - r\cdot\Delta t)V^j_n = P_u\cdot V^{j+1}_{n+1} + P_m\cdot V^j_{n+1} + P_d\cdot V^{j-1}_{n+1}$$

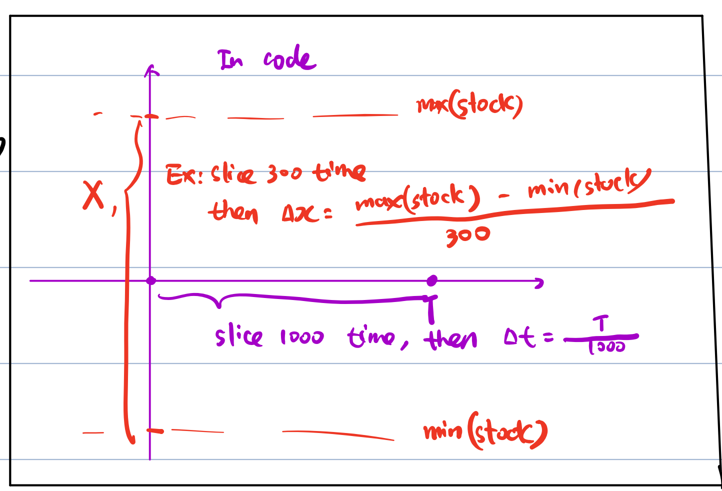
$V^j_n \leftarrow V^j_{n+1}$

$$V^j_n = \underbrace{\frac{1}{1 + r\cdot\Delta t}}_{\text{discount}}\left[P_u V^{j+1}_{n+1} + P_m V^j_{n+1} + P_d V^{j-1}_{n+1}\right]$$

$$P_u = \left[\frac{\sigma^2}{2\Delta x^2} + \frac{r - d - \frac{1}{2}\sigma^2}{2\Delta x}\right]\Delta t$$

$$P_m = 1 - \frac{\sigma^2\Delta t}{\Delta x^2}$$

$$P_d = \frac{\Delta t}{2}\sigma^2\cdot\frac{1}{\Delta x^2} - \frac{\Delta t}{2\Delta x}(r - d - \frac{1}{2}\sigma^2)$$



↓

Intuition: The stock price change inside $\Delta t = \sigma\sqrt{\Delta t}$ (to make sure inside the Δt grid, capture the maximum diffusion)

condition on Δt and Δx

from historical or stress period σ.

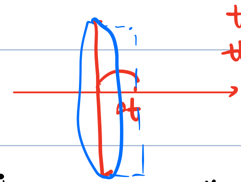
From the note: $\boxed{\Delta x = \sigma_{max}\sqrt{\Delta t}}$

$$\mu = r - q - \frac{\sigma^2}{2}$$

$$p = \frac{\sigma^2 \Delta t}{2(\Delta x)^2} = \frac{\sigma^2}{2\sigma_{max}^2}$$

$$\frac{\sigma^2 \Delta t}{2\sigma_{max}^2 \cancel{\Delta t}} =$$

$$\Rightarrow \begin{cases} p_m = 1 - 2p > 0 \\ p_u = \frac{1}{2}\frac{\sigma^2 \cdot \Delta t}{\Delta x^2} + \frac{\mu \cdot \Delta t}{2\Delta x} = p + \frac{\mu \Delta t}{2\sigma_{max}\cdot\sqrt{\Delta t}} \\ \quad = p + \frac{\mu \cdot \sqrt{\Delta t}}{2\sigma_{max}} \\ p_d = p - \frac{\mu\sqrt{\Delta t}}{2\sigma_{max}} \end{cases}$$



condition need to satisfy

$$\Longrightarrow \begin{cases} p < \frac{1}{2} \Rightarrow \frac{\sigma^2}{2\sigma_{max}^2} < \frac{1}{2} \\ p + \frac{\mu\sqrt{\Delta t}}{2\sigma_{max}} > 0 \\ p - \frac{\mu\sqrt{\Delta t}}{2\sigma_{max}} > 0 \end{cases} \Rightarrow \begin{matrix} -\frac{\mu\sqrt{\Delta t}}{2\sigma_{max}} < p \\ \frac{\mu\sqrt{\Delta t}}{2\sigma_{max}} < p \end{matrix} \Rightarrow \left|\frac{\mu\sqrt{\Delta t}}{2\sigma_{max}}\right| < p$$

$$\left|\frac{\mu\sqrt{\Delta t}}{2\sigma_{max}}\right| < \frac{\sigma^2}{2\sigma_{max}^2}$$

$$|\mu|\sqrt{\Delta t} < \frac{\sigma^2}{\sigma_{max}}$$

$$\frac{|\mu|\sqrt{\Delta t}}{\sigma_{max}} < \frac{\sigma^2}{\sigma_{max}^2} < 1$$

$$\Rightarrow \frac{|\mu|\sqrt{\Delta t}}{\sigma_{max}} < 1$$

$$\Longrightarrow \begin{cases} \frac{\sigma^2}{2\sigma_{max}^2} < \frac{1}{2} \\ \frac{|\mu|\sqrt{\Delta t}}{\sigma_{max}} < 1 \end{cases}$$

satisfied for sure

satisfy this two condition

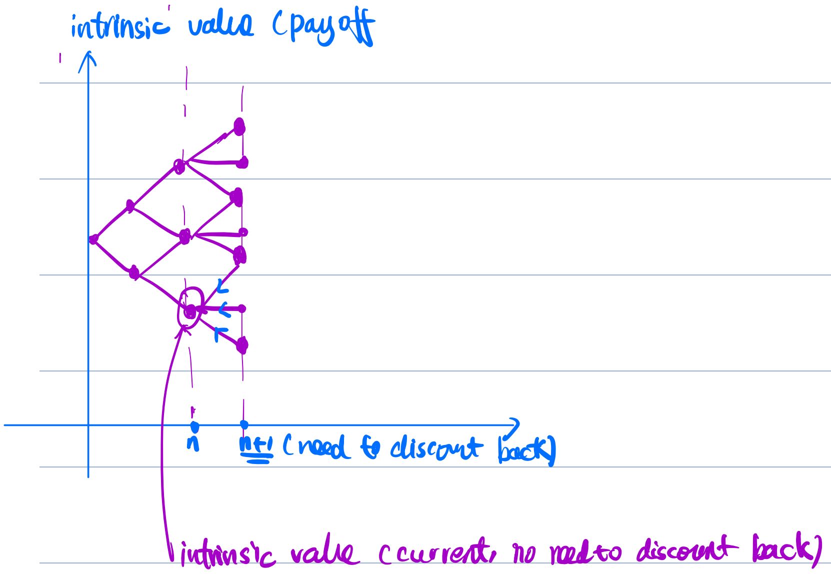
$$|\mu|\sqrt{\Delta t} < \sigma_{max}$$

$$\left(r - q - \frac{\sigma^2}{2}\right)\sqrt{\Delta t} < \sigma_{max}$$

$$\sqrt{\Delta t} < \frac{\sigma_{max}}{r - q - \frac{\sigma^2}{2}}$$

$$\Delta t < \left[\frac{\sigma_{max}}{r - q - \frac{\sigma^2}{2}}\right]^2$$

Code Logic:



American option call ← if disc(n+1) < intrinsic value ⇒ exercise early (choose intrinsic)

if disc(n+1) > intrinsic ⇒ wait until next step (若 all not exercise early, then American = European)

For call, if the p_d is low, then it is high chance to wait until the end to exercise.

The difference between call and put is just the setting